



Generative AI on AWS

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Agenda

Amazon in Romania

What is Generative AI ?

Use cases

Learning resources

Hands-on Lab

Presence in Romania



amazon

The Amazon logo consists of the word "amazon" in a dark blue, lowercase, sans-serif font. Below the text is a curved orange arrow that starts under the 'a' and points towards the 'n'.

aws

The AWS logo consists of the letters "aws" in a dark blue, lowercase, sans-serif font. Below the text is a curved orange arrow that starts under the 'a' and points towards the 's'.

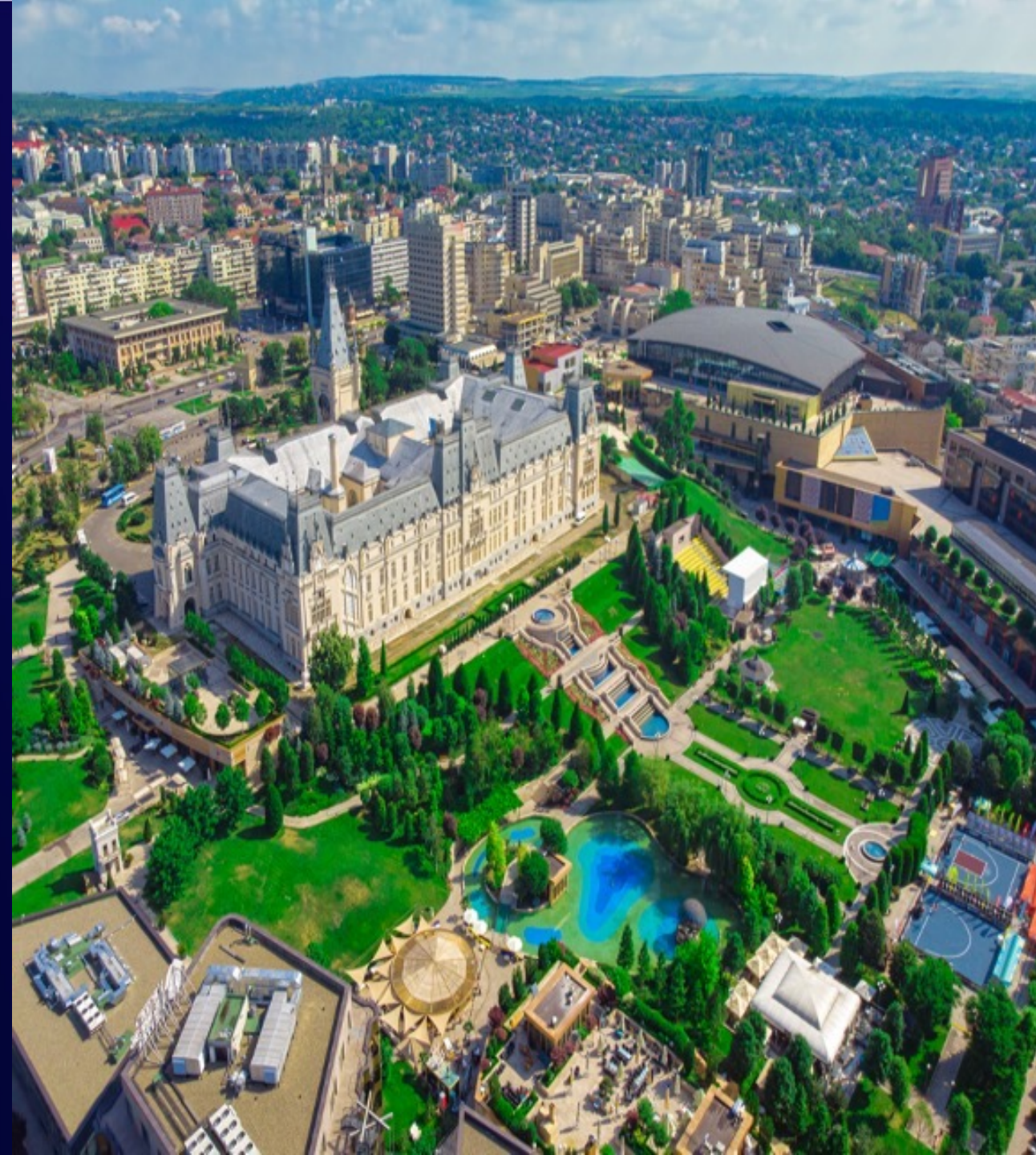
Amazon in Romania

Since 2005 Amazon's
Development Center in Iasi

Was Amazon's first R&D facility
in Eastern Europe

Over 3000+ employees in 3
cities

AWS Office opened in 2021



AWS Bucharest office

Partner and Account Managers,
Solutions Architects and other
supporting roles

Supporting startups, small and
medium-sized businesses,
enterprises, nonprofits, and
government organizations.



Career @ Amazon

<https://www.amazon.jobs/>

Software Dev Engineer

Applied Scientist - Intern 2026

Software Dev Engineer – Intern 2026



Generative AI



Innovation can **transform industries**



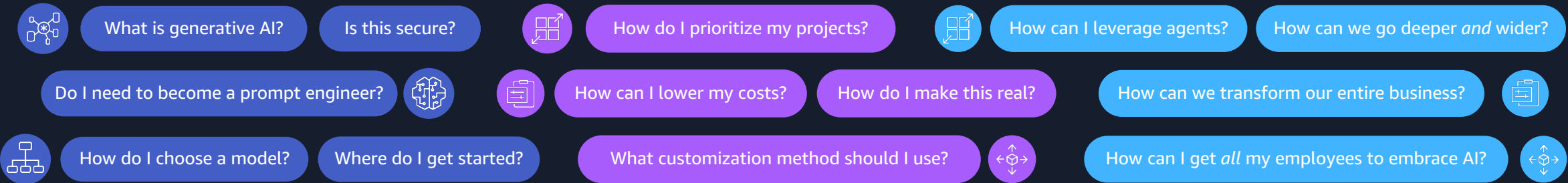
GENERATIVE AI

Gen AI is evolving

2023
The year of POCs

2024
The year of production

2025
The year of business value



How does it work ?



Demystifying common GenAI terms (and why they are important to outcomes)

Vector

Numerical representations of text that saves the meaning and the semantic relationship with other words

Prompt Engineering

Improves accuracy by providing text instructions.

Pre-Training

When you create an Foundation Model from scratch.

Embeddings

Collection of Vectors that belong to a piece of text. Embeddings models transform text into Vectors, that are then stored in Vector DB.

RAG

Retrieval Augmented Generation, uses internal company documentation to find the answers

Re-Training

When you use a FM model and **unlabeled** company data to teach the model about that data so that it can then respond questions about it. Increases accuracy in the answers.

Parameters

Memory of the models, the larger the number of parameters the better it performs
i.e. Llama 3 - 8B, Llama 3.1 405B

Agentic System

Uses the reasoning capabilities of an LLM to automate processes/tasks

Fine Tuning

When you use a FM model and **labelled** company data to teach the model about that data so that it can then respond questions about it. Increases accuracy in the answers.

AI

Machine
learning

Simple

Inputs



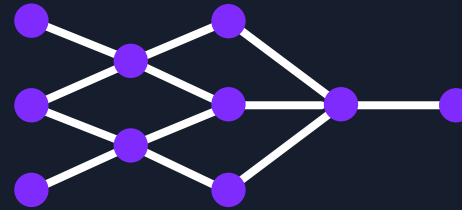
Simple

Outputs

Deep
learning

Complex

Inputs



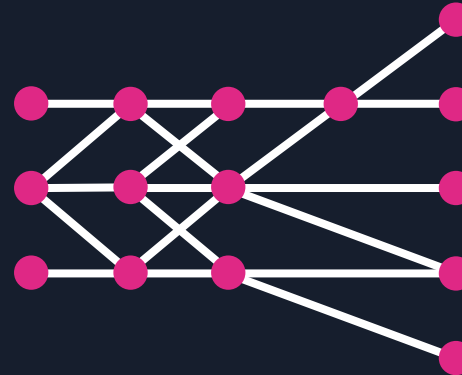
Simple

Outputs

Foundation
models

Complex

Inputs



Complex

Outputs



Generative AI core capabilities

Capabilities



Generation



Question
answering



Summarization



Translation



Correction



Classification

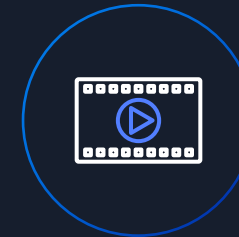
Input/Output



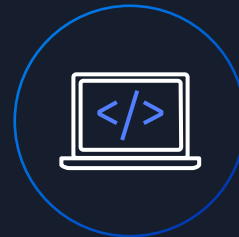
Text



Images



Video/Audio



Code

Generative AI use cases across industries

Enhance customer experiences

Chatbots

Virtual assistants

Conversation analytics

Personalization

Boost employee productivity & creativity

Conversational search

Summarization

Content creation

Code generation

Data to insights

Optimize business processes

Document processing

Data augmentation

Fraud detection

Process optimization

How to implement such use case on AWS ?



Content generation

Request

Product: Smart Watch

Keywords: Health monitoring, notifications, contactless payments, voice assistant, wearable

Task: Write a creative product summary using keywords in 30 words.



Marketing content
writer

Response

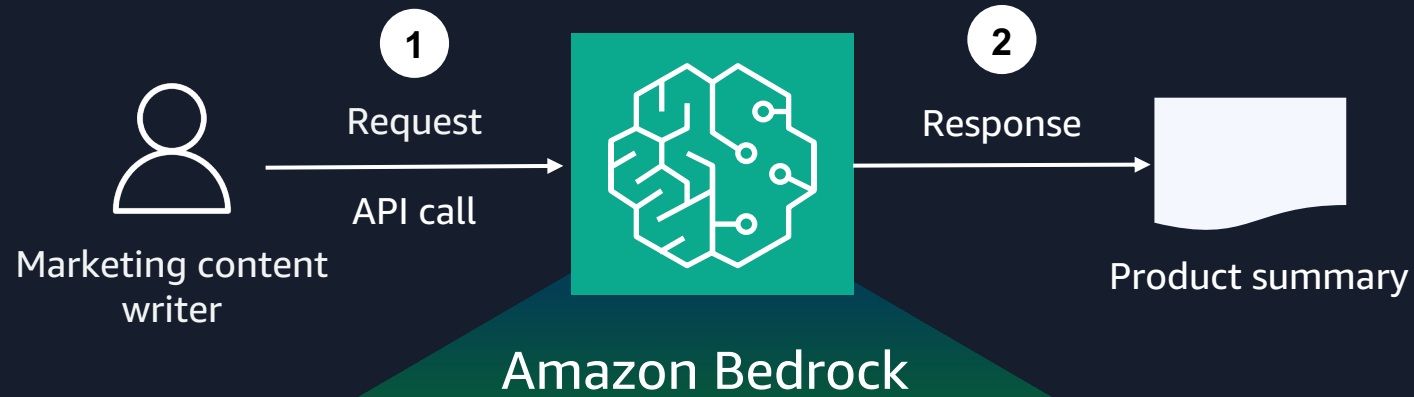
This sleek Smart Watch keeps tabs on your health stats and delivers notifications to your wrist. Make contactless payments and access your voice assistant on the go with this multifunctional wearable



Generative AI
application

Content generation simplified

Task: Write a creative product summary using keywords in 30 words.



Choice of foundation models

AI21labs ANTHROPIC co:here ∞ Meta AI stability.ai **amazon**



Generative AI
Application

Data Foundation

STORAGE

GOVERNANCE
& COMPLIANCE

DATABASES,
ANALYTICS,
& DATA LAKES

DATA
INTEGRATION

Strategies for implementation and their trade-offs



Prompt engineering



What is a prompt?

A **prompt** is the **information** you pass into a **large language model** to elicit a **response**.



Instructions matter

Specificity, clarity, and persuasiveness are important!



You are a physics teacher. Explain quantum mechanics in two phrases.



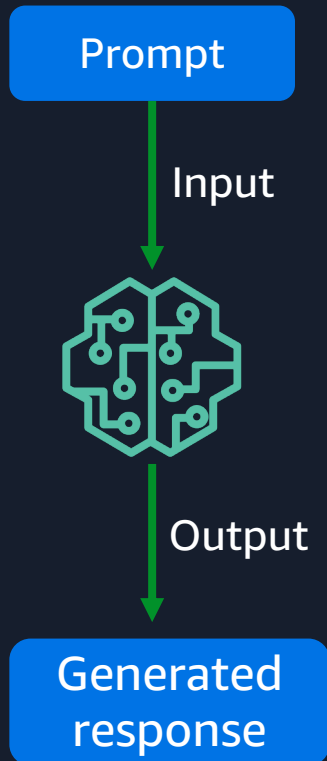
You are a physics teacher. Explain quantum mechanics in two phrases like i was a 5 years old. Make it fun !

Neither humans nor LLMs can read minds.

Prompt engineering tips

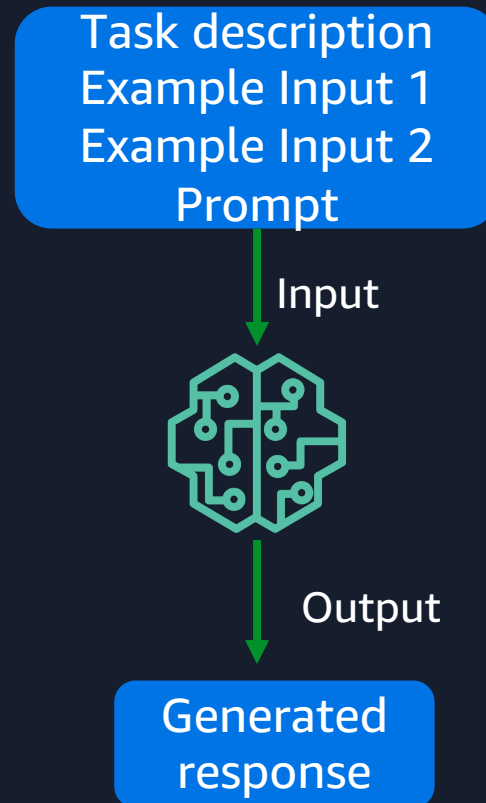
Zero shot prompts

Direct request with sufficient context



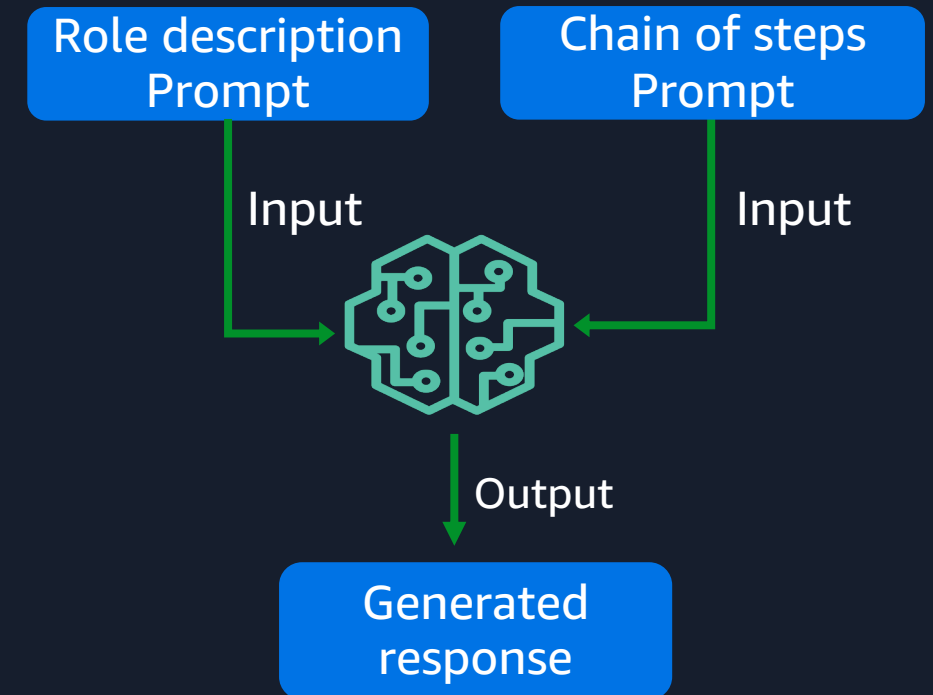
One shot or few shot prompts

Provide one or more examples with a request



Role or Chain of Thought prompts

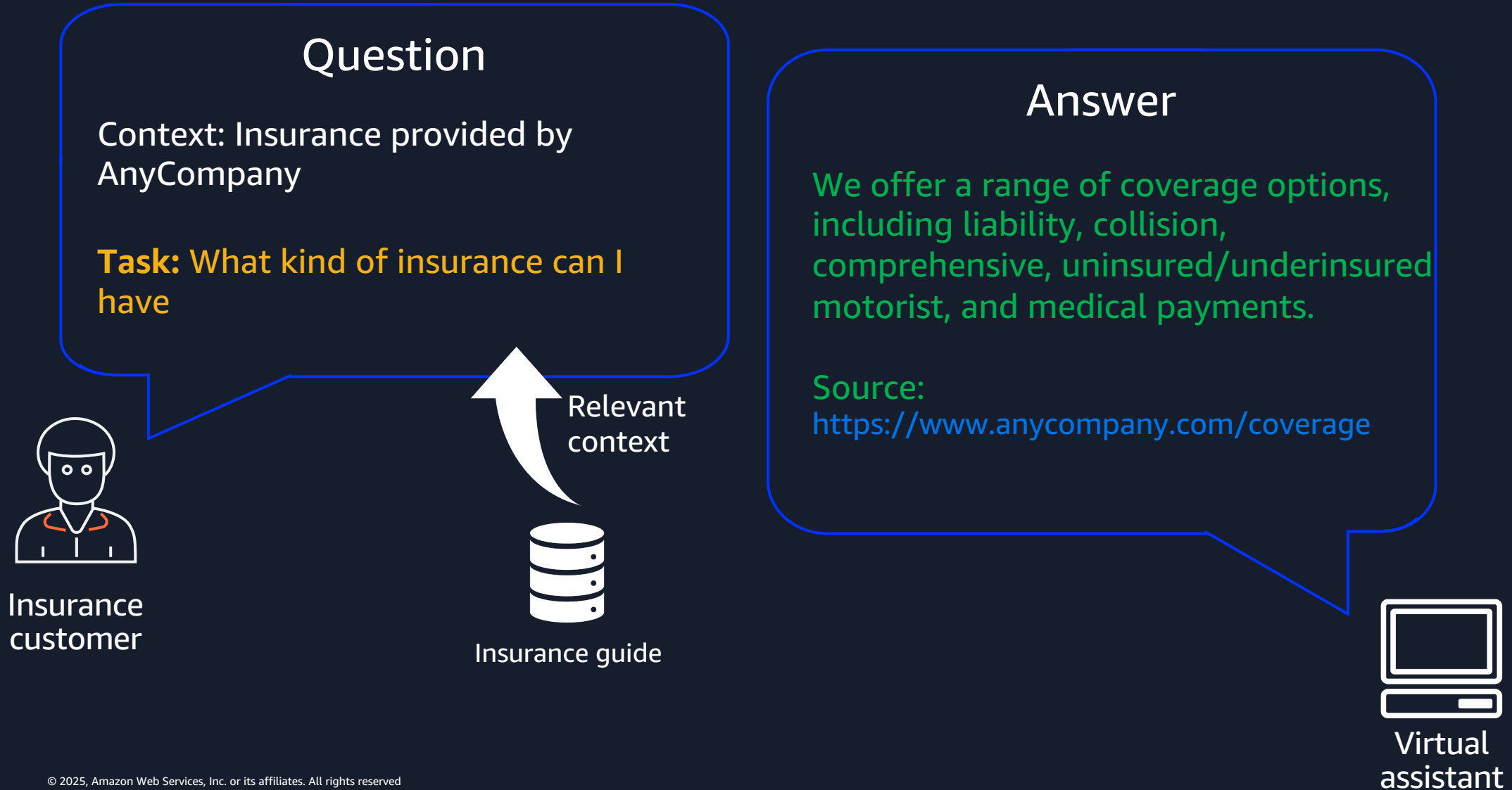
Provide the model with a role or persona for the task
Provide a chain of steps for the model to follow



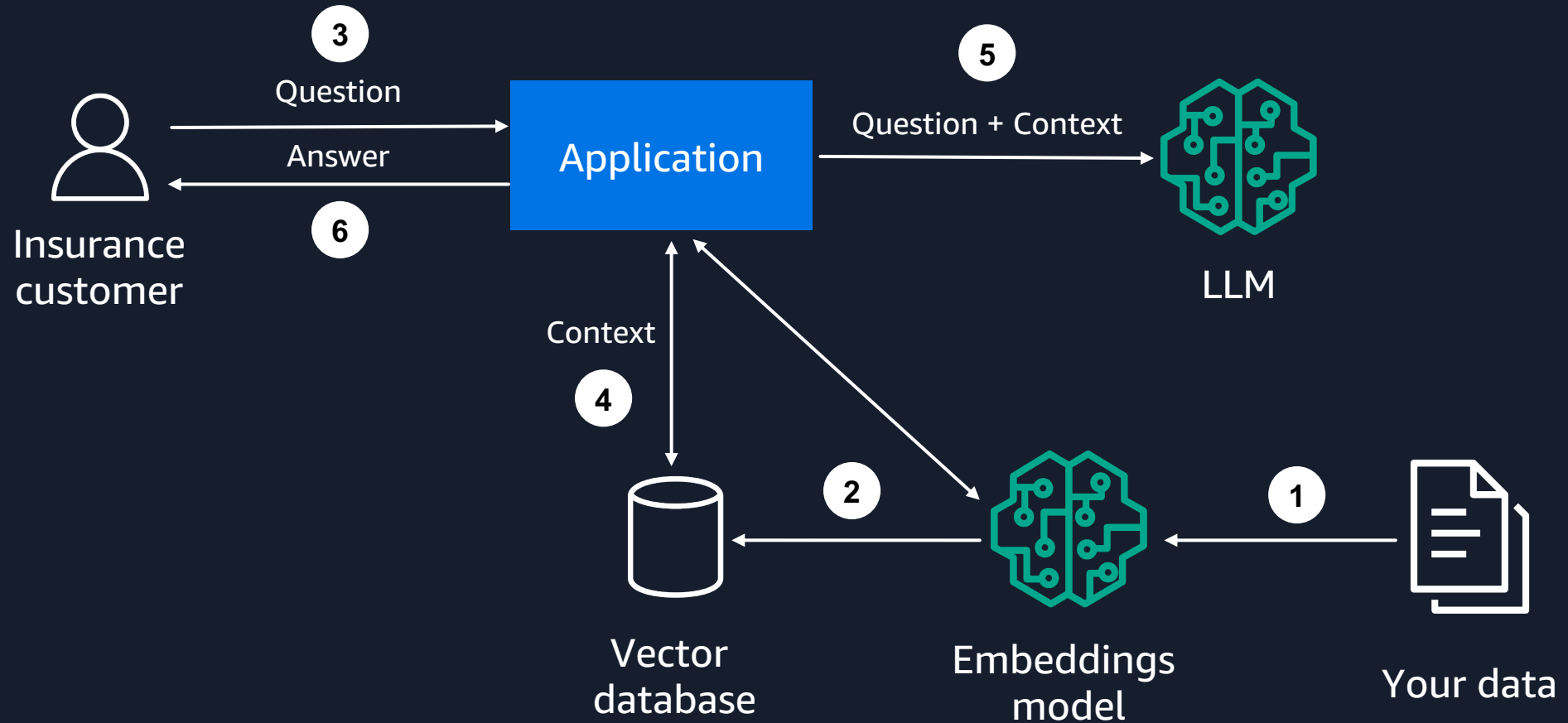
Retrieval Augmented Generation (RAG)



Conversational Q&A



Retrieval Augmented Generation (RAG) concept



What are vector embeddings?

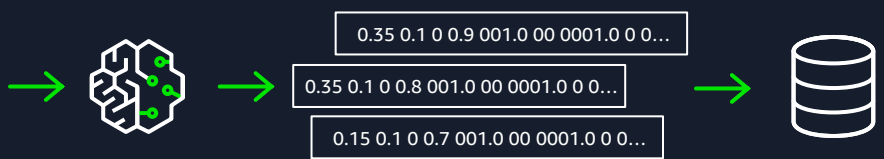


Typical chunks for RAG:

- Words, phrases
- Paragraphs, sections, documents
- Scenes, song sections
- Faces, detected picture elements

Chunking strategies:

- Fixed-size
- Content-aware



Think of it as coordinates:

- 2D vector: [3, 5] - a point on a flat plane (x=3, y=5)
- 3D vector: [3, 5, 2] - a point in 3D space (x=3, y=5, z=2)
- 768D vector: [0.23, -0.45, 0.12, ...] - a point in 768-dimensional space



3D simplified representation. Embeddings used in RAG can have thousands of dimensions, and represent elements that are larger than individual words.

Source: <https://daleonai.com/embeddings-explained>

Agents



Curated knowledge source

Question

Task: What are the medical conditions for patient John100

Task: Generate an email for the next available time slot



Medical staff

Answer

Based on the medical records, the medical conditions John100 has are:

- Type 2 diabetes
- History of hypertension

Next available appointment is on 10 December 2026



Patient Medical History



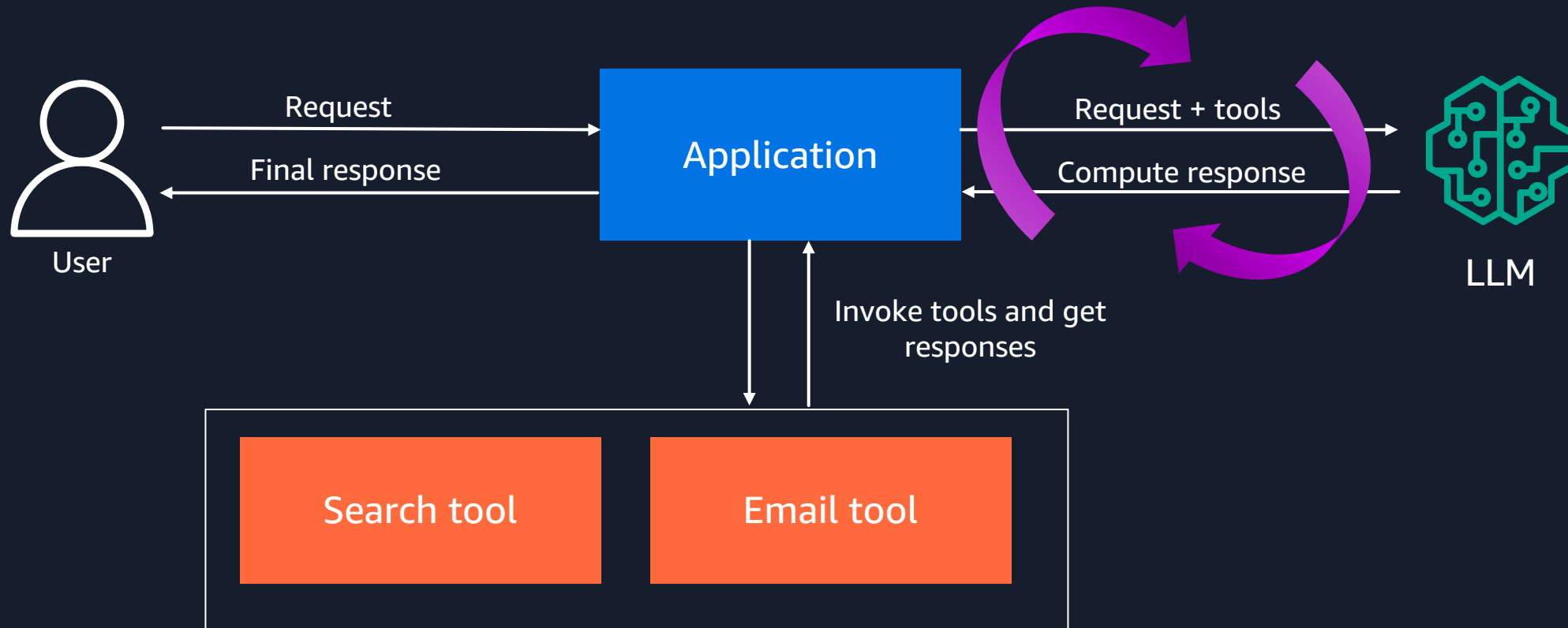
Scheduling System



Medical application

Automated task

Task: Send an email to patient John100 for the next available time slot



Fine-tuning



Fine-tuning example

Ron the dog



Ron the dog wearing a superhero cape



Ron the dog on the moon



Ron the dog in a swimming pool with sunglasses

Smila the cat



Smila the cat on the snow

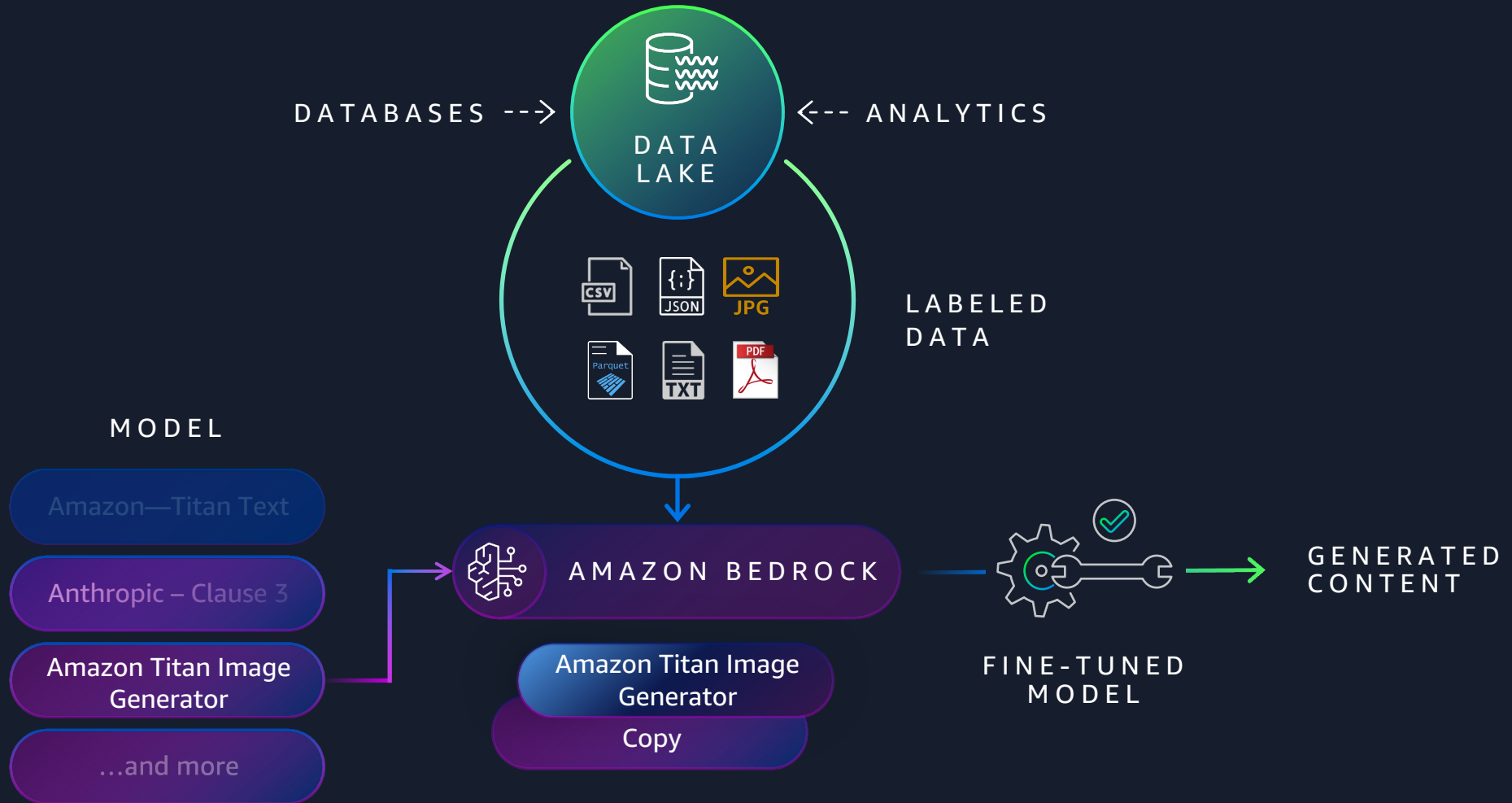


Smila the cat in black and white staring at the camera



Smila the cat wearing a Christmas hat

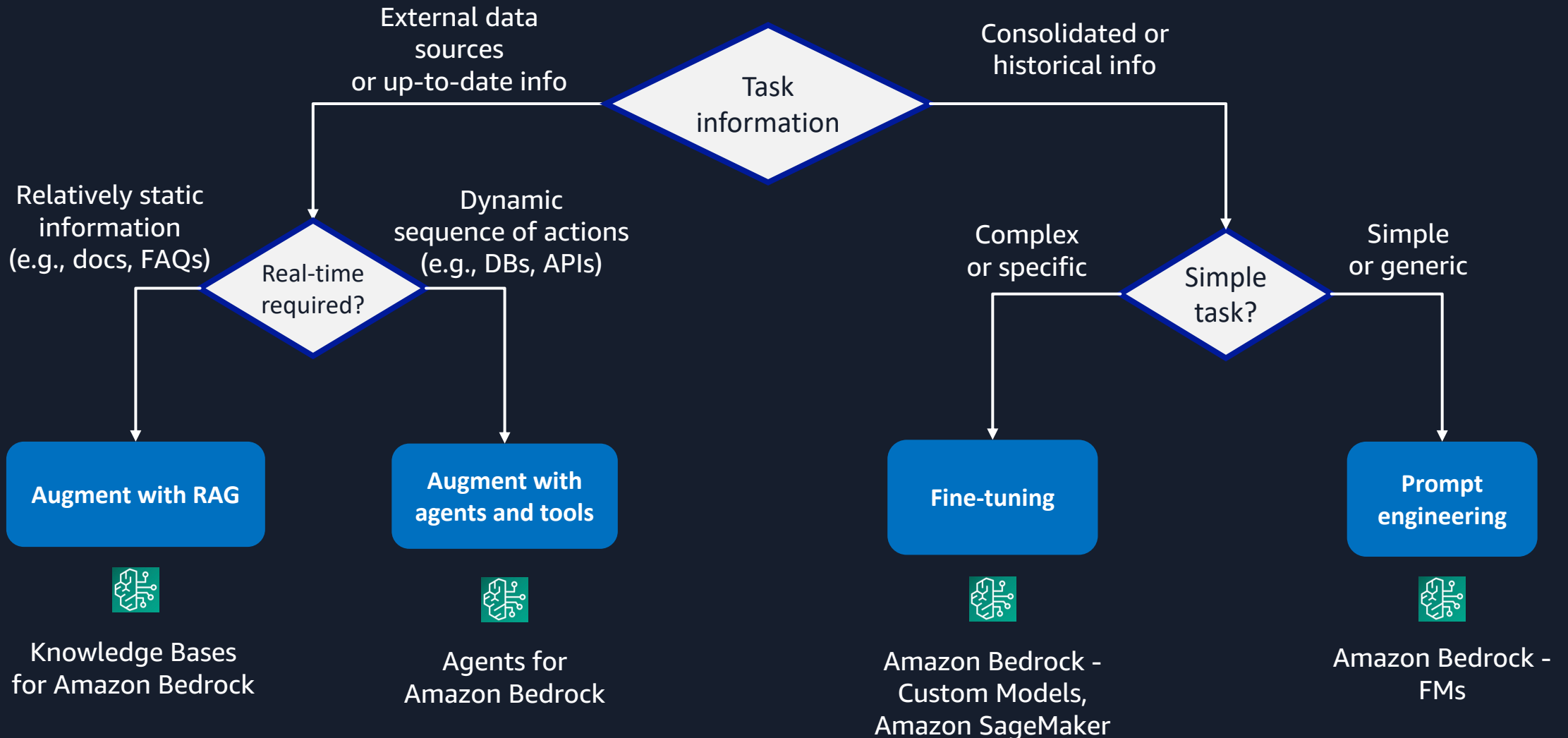
Fine-tuning FMs with your data



Conclusion



Decision Tree



Learn and Be Curios



Learn more ?



AWS Educate offers hundreds of hours of self-paced training and resources for new-to-cloud learners—including hands-on labs.

<https://aws.amazon.com/education/awseducate/>



Join the **AWS Community**.

Students can join or lead an AWS Cloud Club in their region.

<https://community.aws/students>



Validate expertise with an **AWS Certification**.

<https://skillbuilder.aws/>

Accelerate your AI journey with our AWS Ramp-Up Guide for Generative AI

- Recommended course sequencing
- Free digital courses
- Virtual classroom courses
- Whitepapers
- Documentation
- Videos

[Ramp-Up_Guide_Generative_AI.pdf](#)



AWS Ramp-Up Guide: Generative AI

For Business Decision Makers, Developers, Data Scientists and ML Ops Engineers

These are the most salient learning resources from our classrooms, digital curricula, game-based learning, and AWS Lab Environment, that relate to the Generative AI. This Ramp-Up Guide is not intended to be consumed entirely. Different sections tackle different objectives. Some resources will be more relevant to you based on your role. The guide will help you decide where you want to start your learning journey, and which step should be your next.

Share [your opinion](#) of this Ramp-Up Guide with us!

For Business Decision Makers			
Learning Resource		Duration (hrs)	Type
Introduction to Generative AI - Art of the Possible	Fundamental	1	Digital Training
Planning a Generative AI Project	Fundamental	1	Digital Training
Building a Generative AI-Ready Organization	Fundamental	1	Digital Training
Amazon Q Introduction	Fundamental	0.3	Digital Training

For Developers			
Learning Resource		Duration (hrs)	Type
Introduction to Generative AI: Art of the Possible	Fundamental	1	Digital Training
Planning a Generative AI Project	Fundamental	1	Digital Training
Amazon Bedrock Getting Started	Fundamental	1	Digital Training
Foundations of Prompt Engineering	Fundamental	4	Digital Training
Building Generative AI Applications using Amazon Bedrock	Intermediate	4	Digital Training
Amazon Q Developer Getting Started	Fundamental	1	Digital Training
Getting Started with Amazon Kendra	Fundamental	1.5	Digital Training
Getting Started with Amazon Lex	Fundamental	1	Digital Training
\$ Build a question-answering bot using generative AI	Intermediate	1.5	Self-paced Lab
\$ AWS Jam Journey: GenAI reInvent 2023		16	Game-based Learning

Key certifications



Non-IT departments will derive significant value from AI

Employers are willing to pay 39% more to hire AI-skilled workers in sales, marketing, and business operations, while they will pay 37% more for those in finance.¹



“There are a lot AI/ML resources on the internet but not many that address moving ML models to production.” – AWS Partner

Demand for AI and ML Specialists is expected to grow by 40% or 1 million jobs between 2023-27.²



Data is the differentiator when building AI solutions

In 2023, the number of AI jobs postings grew by 119% (44,432 AI jobs) year over year. Job posts for “data engineer” outnumbered those for other job titles.³



Machine learning is the most in-demand AI skill with a high economic value

Workers with AI skills are particularly valuable and command salaries 21% higher than average, but potentially up to 40% higher.⁴

¹ <https://www.zdnet.com/article/workers-with-ai-skills-can-expect-higher-salaries-depending-on-their-role/>

² World Economic Forum, [The Future of Jobs Report 2023](#)

³ OMDIA, [AI Skills Tracker – 1H23 Analysis](#)

⁴ University of Oxford, [Artificial Intelligence Skills Can Increase Salaries by as much as 40%](#)

AWS T&C Badges

AWS Certification



AWS Cloud Quest



AWS Industry Quest



AWS Knowledge



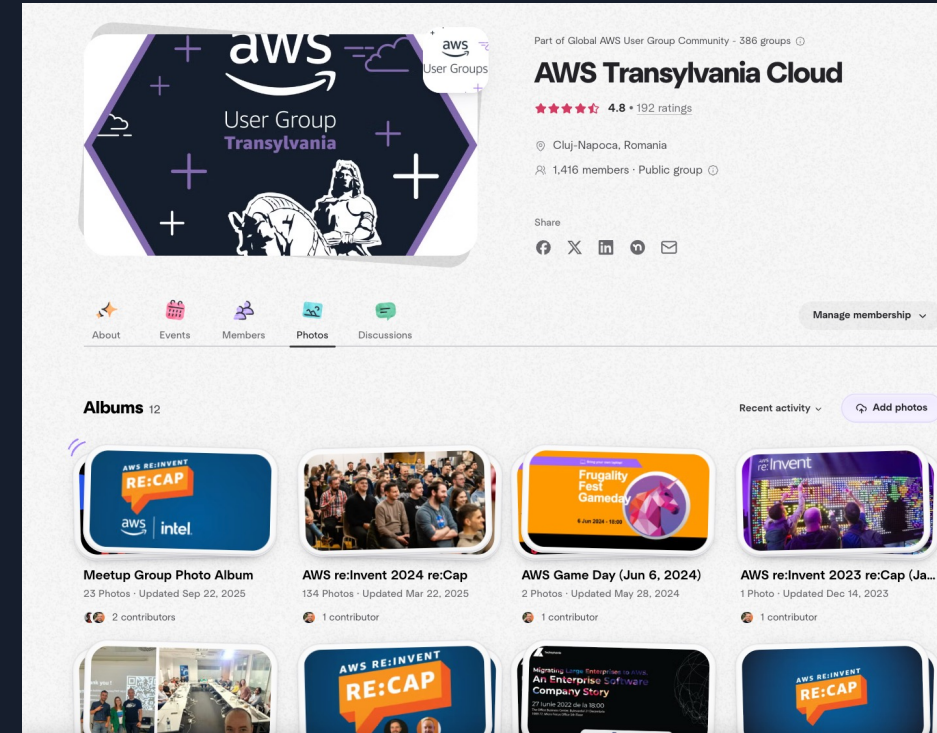
AWS Community

Join or **Start** Student Club



<https://www.meetup.com/aws-cloud-club-at-ustpb/>

Join local AWS Community



<https://www.meetup.com/transylvaniacloud/>



Thank you!

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